

18. Plant Science An experiment was conducted to determine the effect of various levels of phosphorus on the inorganic phosphorus levels in Sudan grass. The data in the table represent the levels of inorganic phosphorus in micromoles (μmol) per gram dry weight. Use the MINITAB output to answer the questions.

Phosphorus Applied, x	Phosphorus in Plant, y
.50 μmol	204 195 247 245
.25 μmol	159 127 95 144
.10 μmol	128 192 84 71

- Plot the data. Do the data appear to exhibit a linear relationship?
- Find the least-squares line relating the plant phosphorus levels y to the amount of phosphorus

applied to the soil x . Graph the least-squares line as a check on your answer.

- Do the data provide sufficient evidence to indicate that the amount of phosphorus present in the plant is linearly related to the amount of phosphorus applied to the soil?
- Estimate the mean amount of phosphorus in the plant if .20 μmol of phosphorus is applied to the soil. Use a 90% confidence interval.

Regression Analysis: y versus x

Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	80.85	22.40	3.61	0.005
x	270.82	68.31	3.96	0.003

Regression Equation

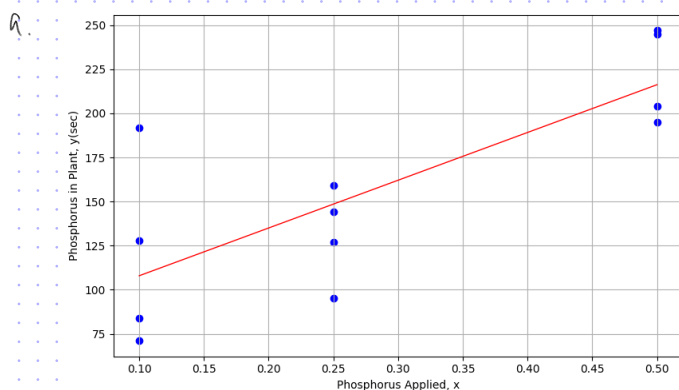
$$y = 80.9 + 270.8x$$

Settings

Variable	Setting
x	0.2

Prediction

Fit	SE Fit	90% CI	90% PI
135.015	12.6264	(112.130, 157.900)	(60.6448, 209.386)



We can observe a positive correlation between phosphorus applied and phosphorus in plant, and some variability within the same x .

b.

$$\hat{y} = a + bx, \quad \bar{y} = 157.5834, \quad \bar{x} = 0.2834$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{88.4667}{0.3267} = 270.82, \quad a = \bar{y} - b\bar{x} = 80.85$$

$$S_{xy} = 88.4667, \quad S_{xx} = 0.3267$$

$$\hat{y} = 80.9 + 270.82x$$

$$c: H_0: \beta = 0, \quad H_a: \beta \neq 0$$

$$\alpha = 0.05$$

$$i) p\text{-value} = 0.003 < 0.05$$

$$ii) \text{Reject } H_0$$

$\Rightarrow$ The data provide sufficient evidence of a linear relationship

$$d. 90\% \text{ CI} = (112.130, 157.900)$$

$$t_{0.05, 10} = 1.812$$

$$x_0 = 0.2$$

$$SE_{\hat{y}} = \sqrt{MSE \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} = \sqrt{5 \left(\frac{1}{11} + \frac{(0.2 - 0.2834)^2}{0.3267} \right)} = 12.6264$$

$$s = 39.0619$$